

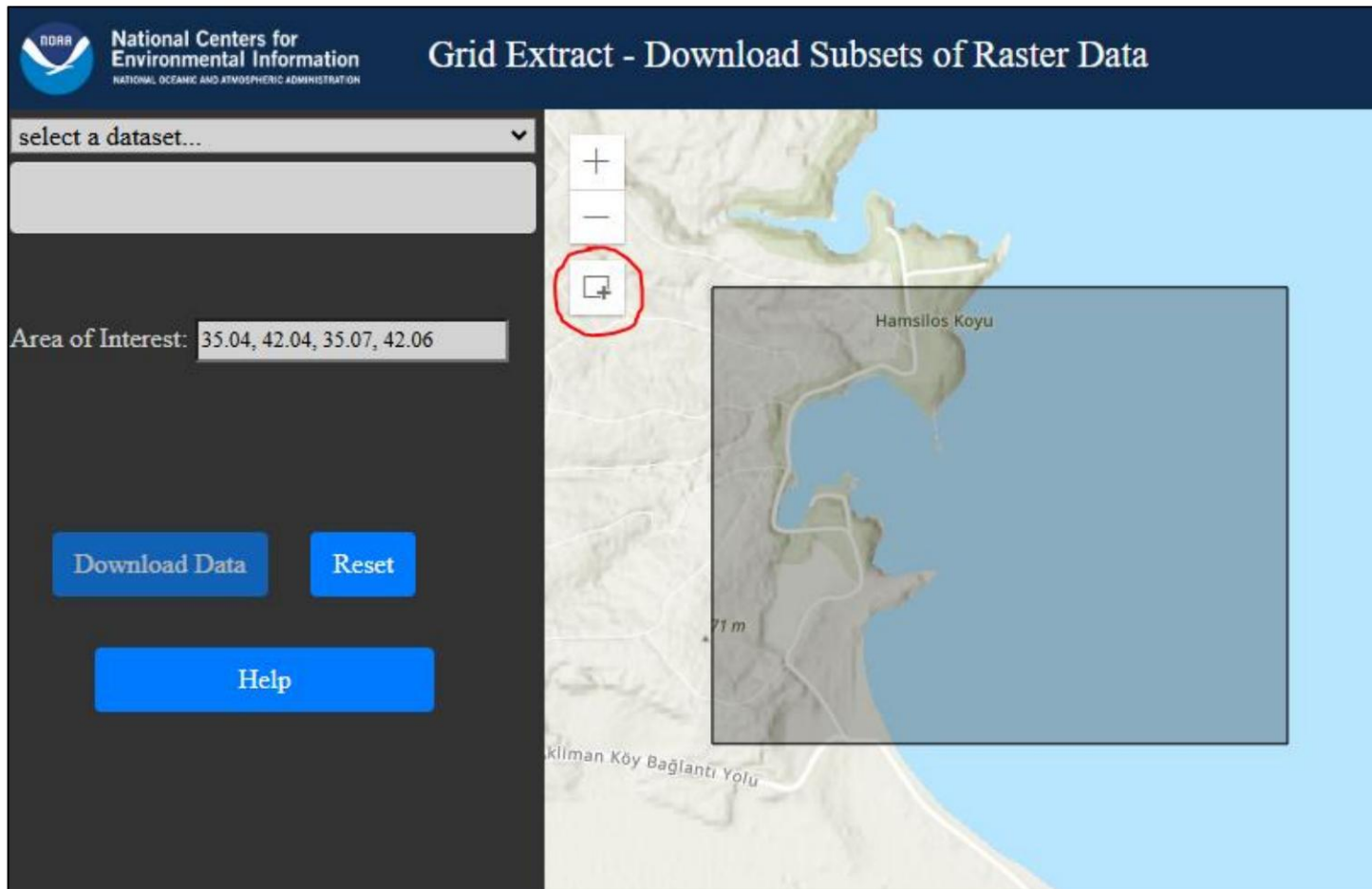
HOW TO DOWNLOAD ERA5 WIND DATA ?

- ERA5 is the latest climate *reassessment* produced by ECMWF.
and, along with uncertainty estimates, many atmospheric, land surface and
It provides hourly data on sea condition parameters.
- (ECMWF: The European Center for Medium-Range Weather
Forecasts –
(European Centre for Medium-Range Weather Forecasts)

First, you need to know the coordinates of the area where you will be working. The coordinates of your region.

You can use the following website or Google Earth to find it.

<https://www.ncei.noaa.gov/maps/grid-extract/>



Inside the red circle

After clicking the icon, select your region.

You can put it inside a frame.

Enter your coordinates on the left side of the screen.

found "Area of Interest"

You can see it in the box.

MEMBERSHIP PROCEDURES

To download ERA5 data, you need to register. There are two websites you can use to register. The relevant links are:

<https://cds.climate.copernicus.eu/> or

<https://www.ecmwf.int/>

When you access these sites, you need to click the **Login – Register** button in **the UPPER RIGHT corner** .

You can register using either your school email address or your personal email address. Data download process...

Since we will be doing this through the "Copernicus" website, you must log in from there.

DATA DOWNLOAD PROCESS

To download the data, after logging into the "Copernicus" website, go to the following link:

<https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels?tab=download>

The screenshot shows the Copernicus Climate Data Store (CDS) website. At the top, there are logos for the European Union, Copernicus, and ECMWF. The user is logged in as SERCAN CIVELEK. The main navigation bar includes links for Climate Data Store, Datasets, Applications, User guide, Live, and Background. A 'Your requests' button is also present. A banner message dated 28 Sep 2024 welcomes users to the new CDS system, noting it is in its early days and undergoing enhancements. The main heading is 'ERA5 hourly data on single levels from 1940 to present'. Below this, there are tabs for Overview, Download (which is active), and Documentation. A warning box dated 10 Oct 2024 states that the final validated ERA5 differs from ERA5T from July 2024 until further notice, with a link to a forum announcement. A 'References' section includes a link for 'Citation and attribution' and a DOI: 10.24381/cds.adbb2d47.

This site has been active since 1940.

up to the present day

meteorological data

It is receiving.

To start the download

Follow the steps.

STEP 1: Select Product Type ÿ Reanalysis

Product type [Select all](#) [Clear all](#)

☒ Reanalysis ☐ Ensemble members

☐ Ensemble mean ☐ Ensemble spread

STEP 2: Click on Variable ÿ «Ocean Waves».

Under the Variables option, you can select the type of data you want to download. For wave analysis, see offshore data.

You need to download the wind data: Click on the "Ocean Waves" option.

(Go to the next slide for the image)

ATTENTION! Selecting the "Wind" heading here would be an incorrect choice. The wind we want to download is...

We will use this data in the JONSWAP analysis, that is, when determining wave height. In this case, we need to download the WIND DATA OVER THE SEA.

"Under the heading 'Ocean Waves,' many options will open up. These include:"

- «*Ocean surface stress equivalent 10m neutral wind direction*» and
- Select the options "*Ocean surface stress equivalent 10m neutral wind speed*" .

Climate Data Store

DatasetsApplicationsUser guideLiveBackground

Ocean waves

Select allClear all

☐ Air density over the oceans

☐ Mean direction of total swell

☐ Mean square slope of waves

☐ Mean wave direction of third swell partition

☐ Mean wave period based on first moment for wind waves

☐ Mean wave period of second swell partition

☐ Normalized energy flux into ocean

☒ Ocean surface stress equivalent 10m neutral wind speed

☐ Significant height of total swell

☐ Significant wave height of third swell partition

☐ Wave spectral kurtosis

☐ Coefficient of drag with waves

☐ Mean direction of wind waves

☐ Mean wave direction

☐ Mean wave period

☐ Mean wave period based on second moment for swell

☐ Mean wave period of third swell partition

☐ Normalized energy flux into waves

☐ Peak wave period

☐ Significant height of wind waves

☐ Wave spectral directional width

☐ Wave spectral peakedness

☐ Free convective velocity over the oceans

☐ Mean period of total swell

☐ Mean wave direction of first swell partition

☐ Mean wave period based on first moment

☐ Mean wave period based on second moment for wind waves

☐ Mean zero-crossing wave period

☐ Normalized stress into ocean

☐ Period corresponding to maximum individual wave height

☐ Significant wave height of first swell partition

☐ Wave spectral directional width for swell

☐ Wave spectral skewness

☐ Maximum individual wave height

☐ Mean period of wind waves

☐ Mean wave direction of second swell partition

☐ Mean wave period based on first moment for swell

☐ Mean wave period of first swell partition

☐ Model bathymetry

☒ Ocean surface stress equivalent 10m neutral wind direction

☐ Significant height of combined wind waves and swell

☐ Significant wave height of second swell partition

☐ Wave spectral directional width for wind waves

Request validation

Request size

Product type

Variable

Year

Month

Day

Time

Geographical area

Whole available region

Sub-region extraction

Data format

Download format

OTHER
OPTIONS
MAKE
SURE IT'S
CLOSED!

STEP 3: Selecting the Year, Month, Day, and Time

It is recommended that you download your data **SEPARATELY** for each year . Downloading all years at once is not recommended.

If you select this option, the times (month-day-hour) will be shifted in the file you download.

ATTENTION! It is recommended to download at least 20 years' worth of data. When downloading data, sort from newest to oldest.

The right path must be taken.

- i. Select a single year – Download at least 20 years of data, starting from the most recent (e.g., 2024).
- ii. Select all months ÿ Click on [Select All](#)
- iii. Select all days ÿ Click on [Select All](#)
- iv. Select all the hours ÿ Click on [Select All](#)

(We will be using hourly data in our analyses, so all hours must be selected.)

(Go to the next slide for the image)

Year [Select all](#) [Clear all](#)

- | | | | | | | | |
|-------------------------------|-------------------------------|-------------------------------|-------------------------------|--|-------------------------------|-------------------------------|-------------------------------|
| ☐ 1940 | ☐ 1941 | ☐ 1942 | ☐ 1943 | ☐ 1944 | ☐ 1945 | ☐ 1946 | ☐ 1947 |
| ☐ 1948 | ☐ 1949 | ☐ 1950 | ☐ 1951 | ☐ 1952 | ☐ 1953 | ☐ 1954 | ☐ 1955 |
| ☐ 1956 | ☐ 1957 | ☐ 1958 | ☐ 1959 | ☐ 1960 | ☐ 1961 | ☐ 1962 | ☐ 1963 |
| ☐ 1964 | ☐ 1965 | ☐ 1966 | ☐ 1967 | ☐ 1968 | ☐ 1969 | ☐ 1970 | ☐ 1971 |
| ☐ 1972 | ☐ 1973 | ☐ 1974 | ☐ 1975 | ☐ 1976 | ☐ 1977 | ☐ 1978 | ☐ 1979 |
| ☐ 1980 | ☐ 1981 | ☐ 1982 | ☐ 1983 | ☐ 1984 | ☐ 1985 | ☐ 1986 | ☐ 1987 |
| ☐ 1988 | ☐ 1989 | ☐ 1990 | ☐ 1991 | ☐ 1992 | ☐ 1993 | ☐ 1994 | ☐ 1995 |
| ☐ 1996 | ☐ 1997 | ☐ 1998 | ☐ 1999 | ☐ 2000 | ☐ 2001 | ☐ 2002 | ☐ 2003 |
| ☐ 2004 | ☐ 2005 | ☐ 2006 | ☐ 2007 | ☐ 2008 | ☐ 2009 | ☐ 2010 | ☐ 2011 |
| ☐ 2012 | ☐ 2013 | ☐ 2014 | ☐ 2015 | ☐ 2016 | ☐ 2017 | ☐ 2018 | ☐ 2019 |
| ☐ 2020 | ☐ 2021 | ☐ 2022 | ☐ 2023 | ☒ 2024 | | | |

Under the **"Month"** heading ,
"December" does
not appear to have been selected.
This is NOT a mistake!

The current date this slide
was prepared: November 2024

Therefore, it is not possible
to have any data for
December.

Select All for other years

When you click the button, you
will see that all 12 months are
selected.

Month [Clear all](#)

- | | | | | | |
|---|--|---|---|--|--|
| ☒ January | ☒ February | ☒ March | ☒ April | ☒ May | ☒ June |
| ☒ July | ☒ August | ☒ September | ☒ October | ☒ November | ☐ December |

Day [Clear all](#)

☒ 01	☒ 02	☒ 03	☒ 04	☒ 05	☒ 06	☒ 07	☒ 08
☒ 09	☒ 10	☒ 11	☒ 12	☒ 13	☒ 14	☒ 15	☒ 16
☒ 17	☒ 18	☒ 19	☒ 20	☒ 21	☒ 22	☒ 23	☒ 24
☒ 25	☒ 26	☒ 27	☒ 28	☒ 29	☒ 30	☒ 31	

Time [Clear all](#)

☒ 00:00	☒ 01:00	☒ 02:00	☒ 03:00	☒ 04:00	☒ 05:00
☒ 06:00	☒ 07:00	☒ 08:00	☒ 09:00	☒ 10:00	☒ 11:00
☒ 12:00	☒ 13:00	☒ 14:00	☒ 15:00	☒ 16:00	☒ 17:00
☒ 18:00	☒ 19:00	☒ 20:00	☒ 21:00	☒ 22:00	☒ 23:00

MAKE SURE ALL THE BOXES ARE CHECKED!

STEP 4: Entering the Coordinates

Enter the coordinates of your defined study area under the heading "**Geographical Area**" and then "***Sub-Region Extraction***".

Type after clicking the option.

Geographical area

☐ Whole available region
With this option selected the entire available area will be provided

☒ Sub-region extraction

North

42.06

West

35.04

East

35.07

South

42.04

STEP 5: Data Format ÷ Select «**GRIB**»

STEP 6: Download Format ÷ «**Unarchived (note**

Select » ***zipped if single file***)

- On the right side, under the heading "**Request Validation**," you can find information about your order.

You can check if anything is missing. If you have made a selection...

A green checkmark will appear next to the relevant heading . If this mark is not present on all of them...

You cannot place an order. Please check again!

- You need to repeat all the steps for each order.
- Once everything is ready, click the SUBMIT FORM option.

You can place your order.

- Data availability period of 1 year – 12 months – 31 days and 24 hours.

It takes approximately 20-25 minutes. In some years, this period can last for hours.

Keep following the "IN PROGRESS" message!

(Go to the next slide for the image)

Request validation

Request size

Product type	✓
Variable	✓
Year	✓
Month	✓
Day	✓
Time	✓
Geographical area	
Whole available region	
Sub-region extraction	
Data format	✓
Download format	✓

Product

Submission

End

Status

Actions



ERA5 hourly data on single levels from 1940 to present

2024-11-19

In progress

Check status



03:00:50am

Your request

Request ID

9e253ec8-d5d4-479c-9cda-5ee2ee17a935 [Open request form](#)

__part_of_request_under_embargo

121411793136888416239875392951139979027

Product type



Reanalysis

Variable



Ocean surface stress equivalent 10m neutral wind direction, Ocean surface stress equivalent 10m neutral wind speed

Year



2024

Month



January, February, March, April, May, June, July, August, September, October, November

Day



01, 02, 03, 04, 05, 06, 07, 08, 09, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31

Time



00:00, 01:00, 02:00, 03:00, 04:00, 05:00, 06:00, 07:00, 08:00, 09:00, 10:00, 11:00, 12:00, 13:00, 14:00, 15:00, 16:00, 17:00, 18:00, 19:00, 20:00, 21:00, 22:00, 23:00

Geographical area



North: 35.07°, West: 42.04°, South: 35.04°, East: 42.06°

Data format

GRIB

Download format

Unarchived (not zipped if single file)



- You can view the status of your order in the "Your Requests" section.
- When your order is ready, the status will change from "In Progress" to "Download" . •

When you download, a file like "**4c00f449e0e77624a791310999edd9a1.grib**" will be downloaded. •

The downloaded file has a .grib extension; you need **the PANOPLY** program to convert it to a text file .

- You can use videos on YouTube to help you set up Panoply.

https://www.youtube.com/watch?v=WNs5ZdZMhQY&ab_channel=NASAEarthdata

- To avoid mixing up the text files after creating them, name each one {Year}_Speed.txt and {Year}_Direction.txt
Save it.

ATTENTION! Text files contain COORDINATES. If the area you initially selected is VERY LARGE, you will see more than one different set of coordinates in this text file. Which one is more suitable for you?

If possible, narrow down your area according to those coordinates and place the order again, and be sure to do the same for your other o

Use the same coordinates!

Datasets Catalogs Bookmarks		
Name	Long Name	Type
▼  adaptor.mars.internal-1666807929.0215108-1... adaptor.mars.internal-1666807929.0215108-14296-7-...		Local File
 10_metre_wind_direction height above g... 10 metre wind direction @ Specified height level above ...		Geo2D
 10_metre_wind_speed @ Specified height level above gr...		Geo2D
 height_above_ground		—
 lat		1D
 LatLon_Projection		—
 lon		1D
 reftime		1D
 time		1D

Create Extra Small Plot

Create Small Plot

Create Standard Plot

Create Large Plot

Create Extra Large Plot

Create Jumbo Plot

Create Super Jumbo Plot

Create King Sized Plot

Create Maximum Sized Plot

Export CDL*...

Export CSV...

Export Labeled Text...

Remove Dataset

 10_metre_wind_speed_height_above_ground.txt - Not Defteri

Dosya	Düzen	Biçim	Görünüm	Yardım
time	height_above_ground	lat	lon	10_metre_wind_speed_height_above_ground
0.0	10.0	37.900001525878906	26.400001525878906	4.998047
1.0	10.0	37.900001525878906	26.400001525878906	4.161133
2.0	10.0	37.900001525878906	26.400001525878906	3.4204102
3.0	10.0	37.900001525878906	26.400001525878906	3.2558594
4.0	10.0	37.900001525878906	26.400001525878906	3.586914
5.0	10.0	37.900001525878906	26.400001525878906	4.17334
6.0	10.0	37.900001525878906	26.400001525878906	4.551758
7.0	10.0	37.900001525878906	26.400001525878906	4.961426
8.0	10.0	37.900001525878906	26.400001525878906	5.2543945
9.0	10.0	37.900001525878906	26.400001525878906	5.2973633
10.0	10.0	37.900001525878906	26.400001525878906	5.4033203
11.0	10.0	37.900001525878906	26.400001525878906	5.3408203
12.0	10.0	37.900001525878906	26.400001525878906	5.0444336
13.0	10.0	37.900001525878906	26.400001525878906	4.258301
14.0	10.0	37.900001525878906	26.400001525878906	3.9047852
15.0	10.0	37.900001525878906	26.400001525878906	3.7514648
16.0	10.0	37.900001525878906	26.400001525878906	3.7666016
17.0	10.0	37.900001525878906	26.400001525878906	3.6240234
18.0	10.0	37.900001525878906	26.400001525878906	3.409668

There should only be one Latitude and one Longitude value, as shown in the image!

DETERMINING WIND DIRECTIONS

- The wind direction in the downloaded data is given in degrees. It assumes north is 0 degrees.
- You will need to use a 16-sector wind turbine.
- You need to scan an area of $360 / 16 = 22.5$ degrees. That is, an area of ± 11.25 degrees belonging to the relevant direction.

You need to perform a scan. The resulting directional ranges are as follows:

N (North): 348.75° - 11.25°	S (South): 168.75° - 191.25°
NNE (North-Northeast): 11.25° - 33.75°	SSW (South-Southwest): 191.25° - 213.75°
NE (Northeast): 33.75° - 56.25°	SW (Southwest): 213.75° - 236.25°
ENE (East-Northeast): 56.25° - 78.75°	WSW (West-Southwest): 236.25° - 258.75°
E (East): 78.75° - 101.25°	W (West): 258.75° - 281.25°
ESE (East-Southeast): 101.25° - 123.75°	WNW (West-Northwest): 281.25° - 303.75°
SE (Southeast): 123.75° - 146.25°	NW (Northwest): 303.75° - 326.25°
SSE (South-Southeast): 146.25° - 168.75°	NNW (North-Northwest): 326.25° - 348.75°

EXAMPLE WIND ROSE

After making the classifications, the relevant direction

You obtain **the number of blowings** within the degree range .

To better understand the area covered by the directions

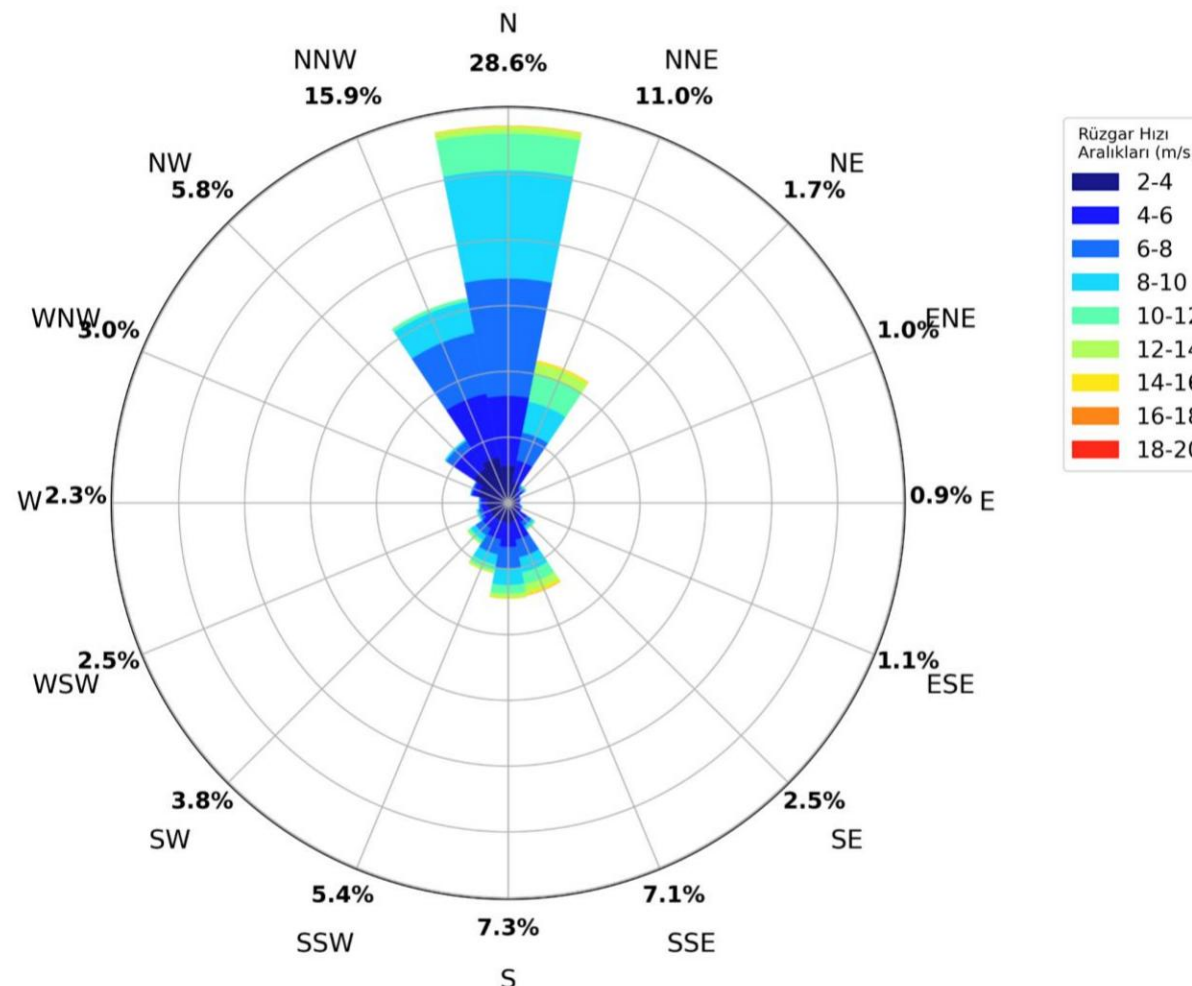
An example pinwheel is shared. Each slice

It scans an area of 22.5 degrees.

You can use this data to build a wind turbine if you wish.

You can create it and for the direction you want.

You can perform your JONSWAP analysis.



For any questions, you can contact my master's student Cem Ali
 Say r: cemaliisagir@gmail.com